

Medical Insight Explorer Agent

Controlled, deterministic, bilingual healthcare claims analytics

v1.2.0 • LangGraph • Gradio

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127.0.0.1:7860

Medical Insight Explorer Agent

Conversational analytics over cleaned Medicare healthcare claims data.
Konversationelle Analyse bereinigter Medicare-Healthcare-Claims-Daten.

Deterministic analytics first. Persona-guided questions. Grounded explanations and optional visual outputs.

Language / Sprache: English

Stakeholder persona / Stakeholder-Persona: Hospital Operations Analyst

Explore provider utilization, claim volume, and reimbursement patterns.

Recommended questions for this persona:

- Compare inpatient and outpatient provider activity
- Show inpatient claims by state
- Show reimbursement distribution

Ask a healthcare analytics question / Frage zur Healthcare-Analyse

Compare inpatient and outpatient provider activity

Run analysis / Analyse ausführen

Clear / Zurücksetzen

Analytical workflow: Multi-step - 2 approved tools executed.

Agent response / Antwort des Agenten

Computed results:

Top inpatient providers:

Provider	claim_count
PRV52019	33
PRV5209	28
PRV52340	25
PRV5300	24
PRV5462	23
PRV53706	22
PRV5231	20
PRV51244	20
PRV51980	18
PRV5300	18

Top outpatient providers:

Provider	claim_count
PRV51409	35
PRV53797	30
PRV51574	37
PRV56011	36
PRV55215	29
PRV51480	31
PRV53818	31
PRV52564	28
PRV54895	26
PRV57306	25

Method note:
This response combines deterministic pandas computations from approved healthcare analytics tools.

Safety note:
This is claims analytics only and does not provide diagnosis, treatment advice, fraud determination, or clinical recommendations.

Visualization output / Visualisierung

Top 10 Inpatient Providers by Claim Count

Provider	Claim Count
PRV52019	33
PRV5209	28
PRV52340	25
PRV5300	24
PRV5462	23
PRV53706	22
PRV5231	20
PRV51244	20
PRV51980	18
PRV5300	18

Supported languages: English | Deutsch

Deterministic analytics • Claims data only • No diagnosis

Use via API • Built with Gradio

Business Problem → Design Response

Why the system is deliberately bounded

The problem

Healthcare claims data is large, relational, and difficult to explore. Natural-language analytics can improve accessibility, but unrestricted AI-generated analysis risks invented calculations and unsupported healthcare conclusions.

The design response

Separate natural-language interaction from numerical authority. Questions may select approved analytical capabilities, while predefined pandas functions calculate the actual healthcare metrics.

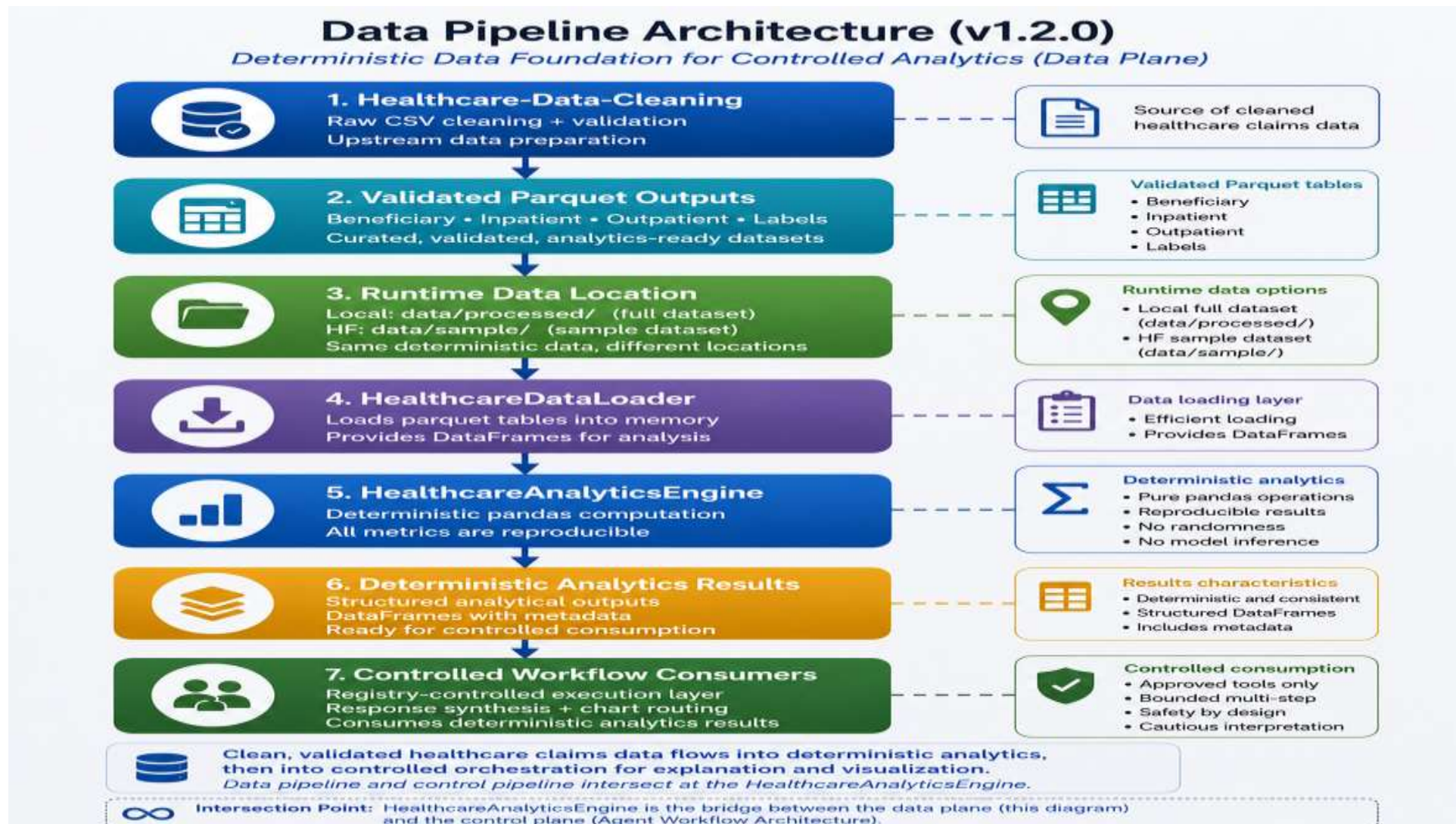
The outcome

A portfolio-grade analytics agent that is reproducible, bilingual, explainable, and intentionally limited to descriptive healthcare claims analysis.

“Natural language may choose approved analytics. It should not invent the analytics themselves.”

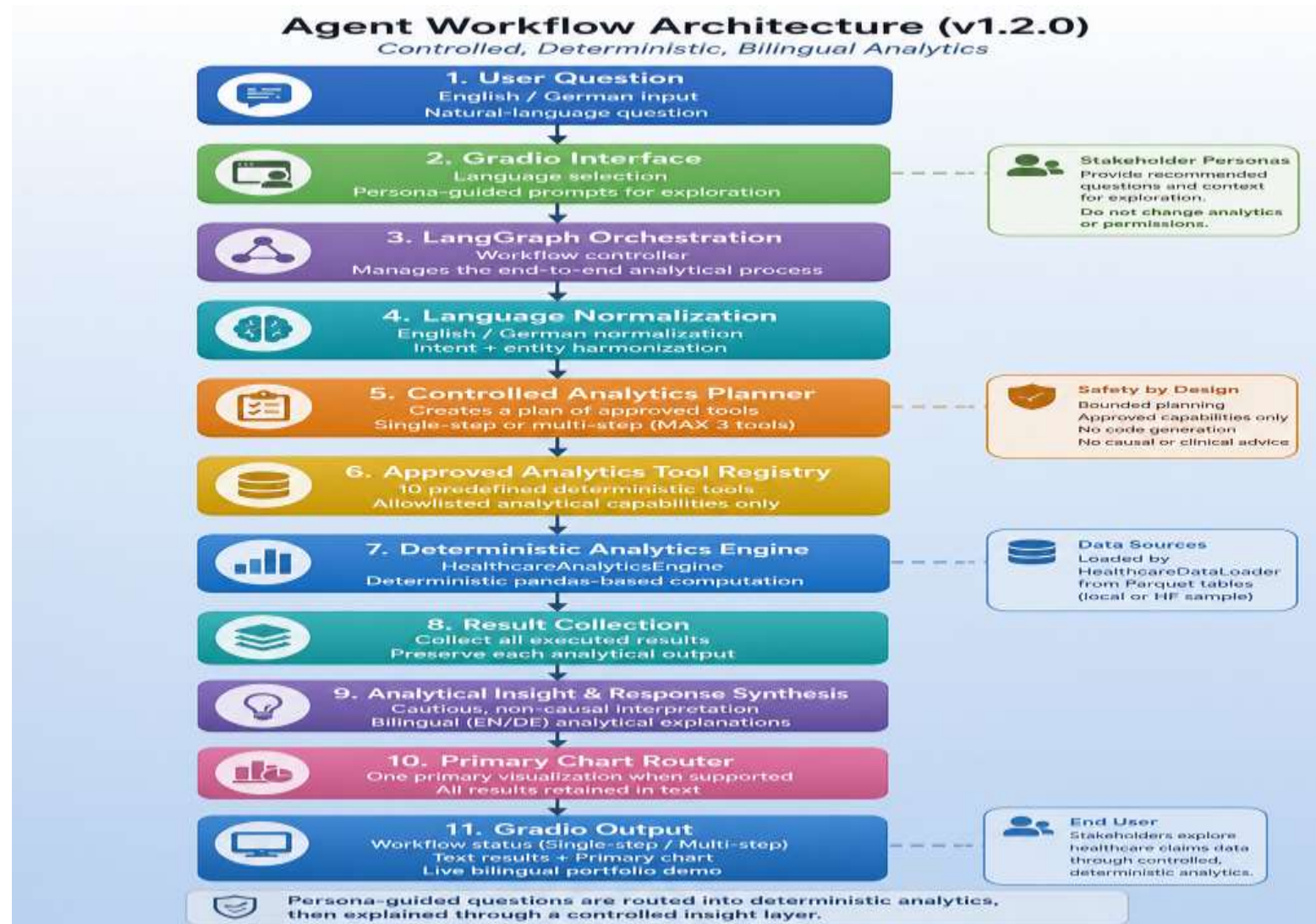
Data Foundation

Upstream cleaning and deterministic analytics remain separate from conversational control



Agent Workflow Architecture

The control plane selects, constrains, executes, synthesizes, and presents



What Changed in v1.2.0

From controlled single-step routing to bounded multi-step orchestration

1. Planner

Creates one-tool plans for simple questions and multi-tool plans only for explicitly supported compound questions.

Maximum: 3 analytics steps.

2. Tool Registry

Ten allowlisted symbolic capabilities form the controller-facing execution boundary.

Unknown or unregistered tools are rejected.

3. LangGraph

Coordinates sequential execution, stores all deterministic results, synthesizes the response, and preserves one primary chart.

Planner selects → Registry constrains → Analytics engine computes → LangGraph coordinates

Approved Analytics Boundary

10 deterministic capabilities; no arbitrary Python, SQL, pandas, or dataframe execution

Dataset

table_shapes
age_summary

Claims

inpatient_summary
outpatient_summary

Providers

top_inpatient_providers
top_outpatient_providers

Geography

inpatient_claims_by_state
outpatient_claims_by_state

Reimbursement

diabetes_cost_summary
reimbursement_distribution

Supported compound workflow

Compare inpatient and outpatient provider activity

Supported compound workflow

Compare inpatient and outpatient claims by state

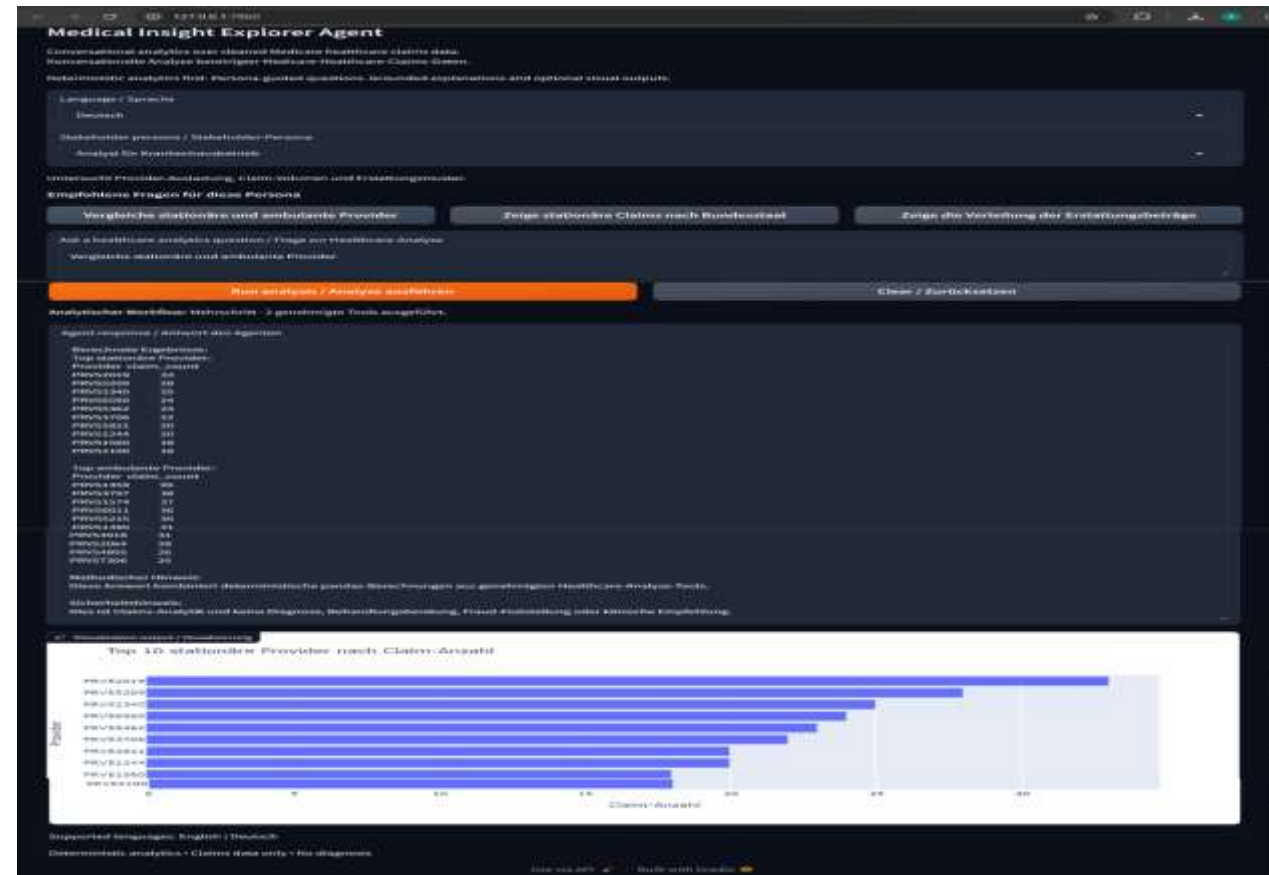
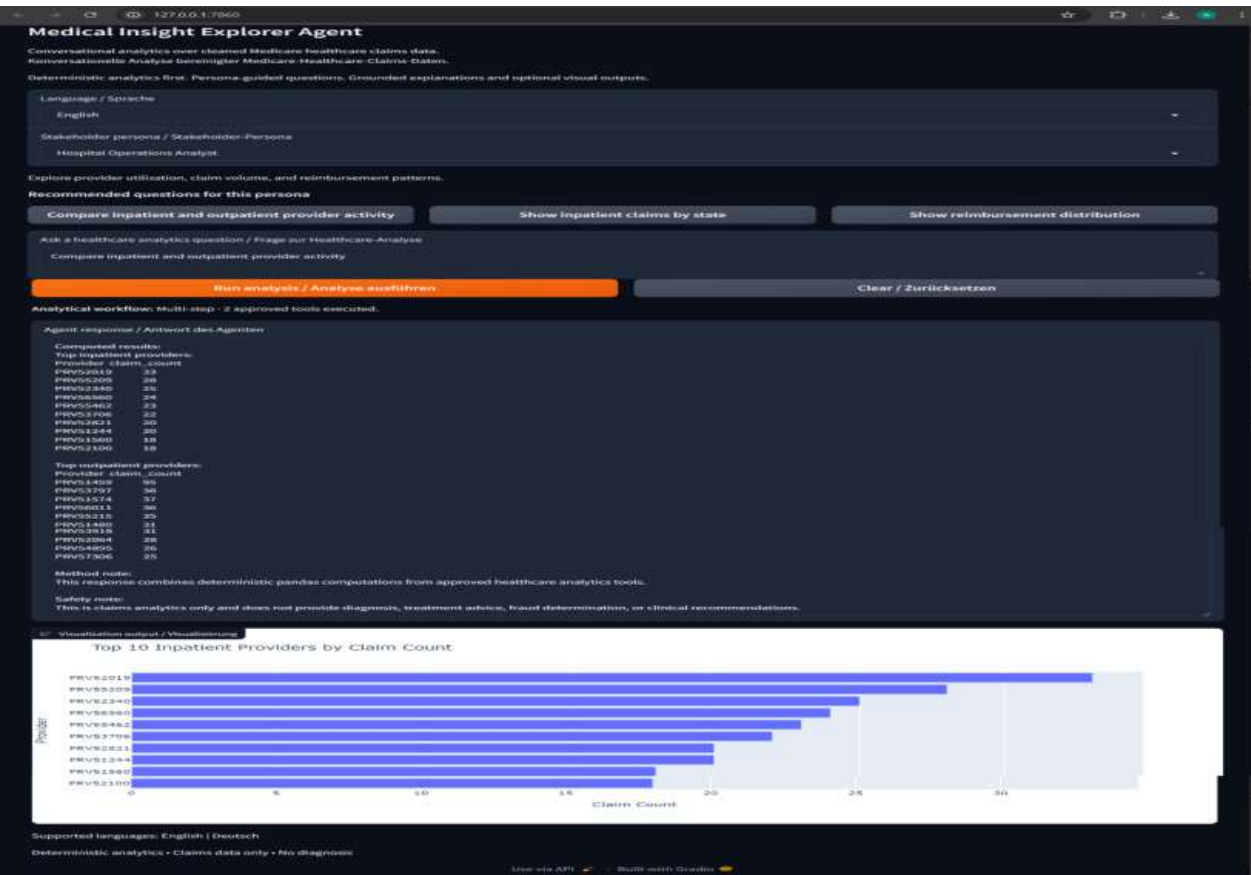
Supported compound workflow

Compare inpatient and outpatient claim summaries

0 tools = unsupported/rejected • 1 tool = single-step • 2–3 tools = bounded multi-step

Bilingual, Persona-Guided UX

Language and stakeholder context improve exploration without changing analytics permissions



English + German

Same deterministic analytical capabilities.

3 stakeholder personas

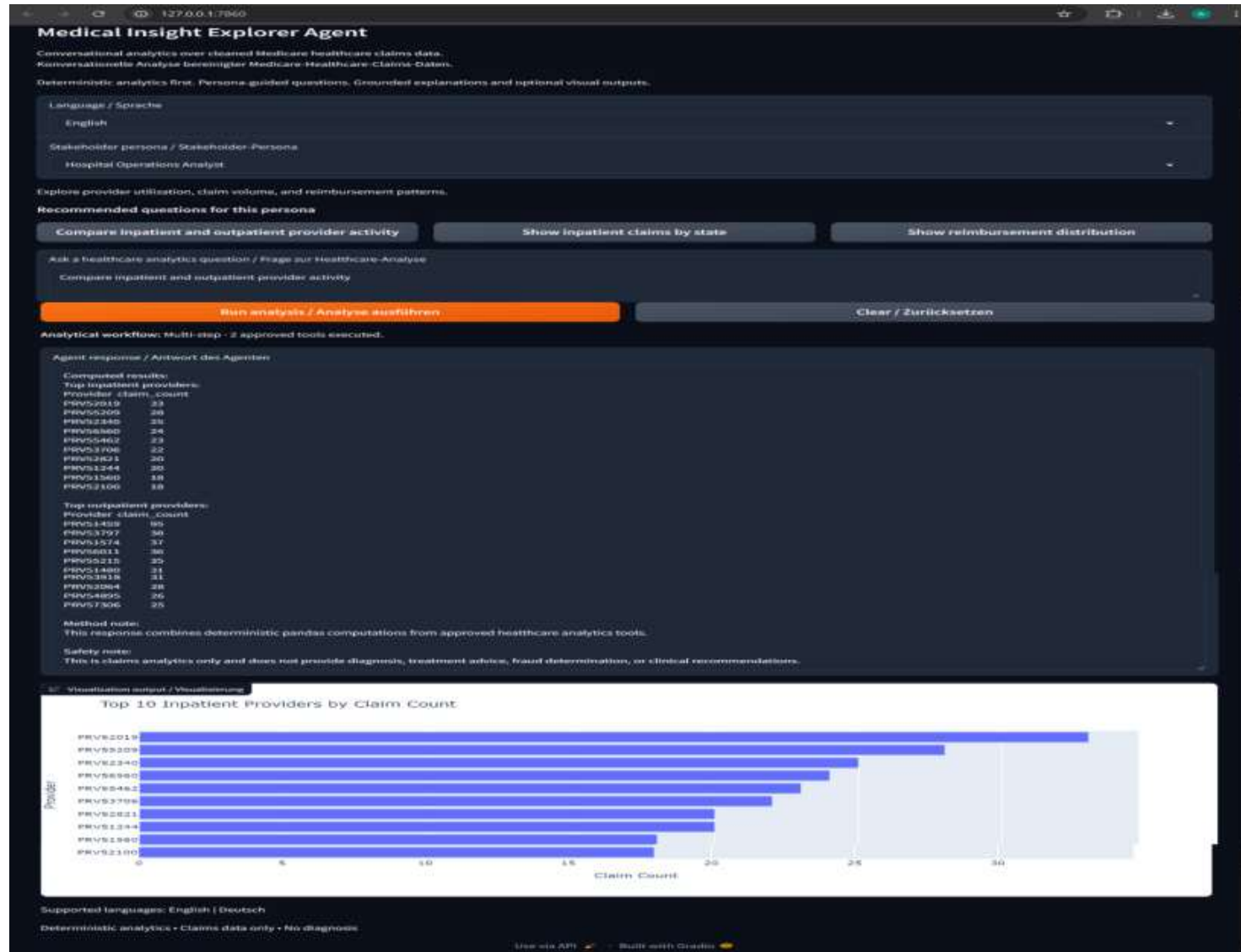
Operations • Fraud investigation • Policy

Persona boundary

Guidance only — no formula or permission changes.

Multi-Step Analytics in the Interface

One question → two approved tools → combined text results → one primary visualization



Visible workflow status

Multi-step · 2 approved tools executed

Multi-result response

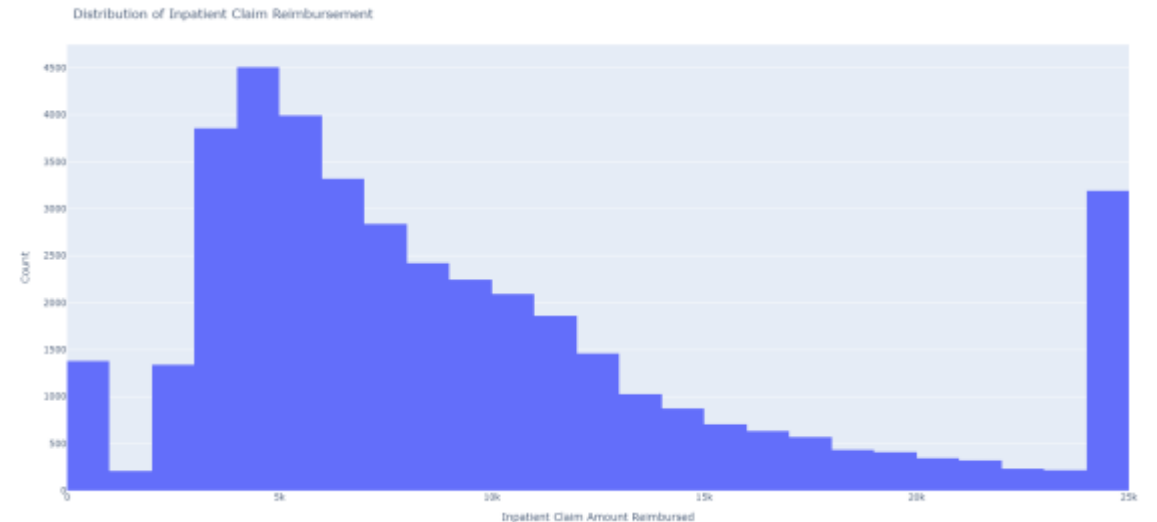
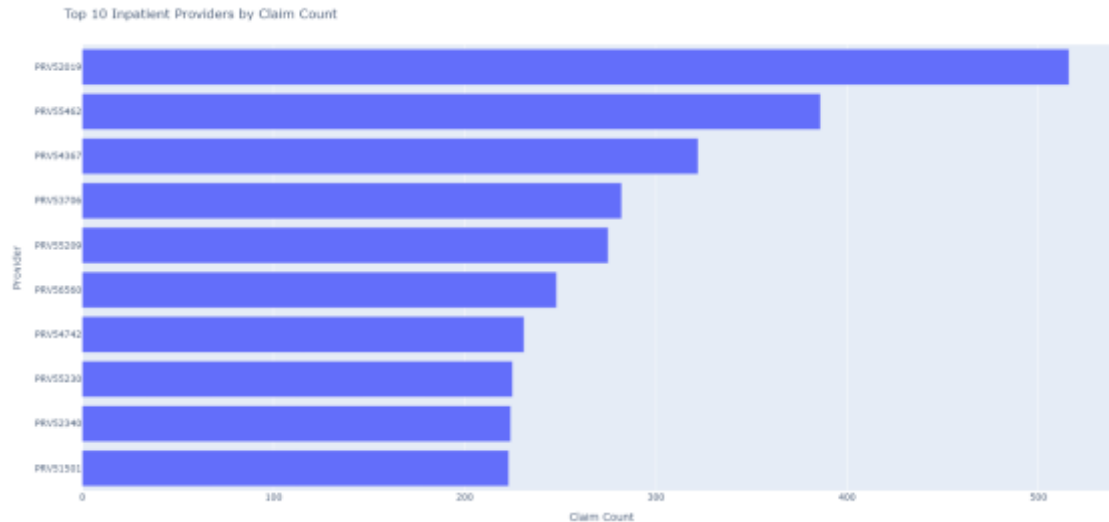
Both inpatient and outpatient provider results are retained in the textual output.

Primary-chart compatibility

The current release intentionally renders one primary chart rather than a multi-chart dashboard.

Representative Analytics Outputs

Deterministic results are calculated first; interpretation is added afterward



Provider activity

Provider claim volume is concentrated among a smaller subset of providers, supporting descriptive utilization monitoring and operational review.

Reimbursement distribution

The distribution is strongly right-skewed: most claims fall in lower-to-mid reimbursement ranges, with a smaller high-cost tail.

Safety & Responsible Analytics

The constraints are architectural, not just wording in the interface

What the system enforces

- ✓ Predefined deterministic pandas calculations
- ✓ Allowlisted controller-facing tool registry
- ✓ Bounded planning: MAX_ANALYTICS_STEPS = 3
- ✓ Unsupported and causal requests can execute zero tools
- ✓ Personas guide exploration only

What the system does not do

- × No diagnosis or treatment recommendations
- × No patient-specific clinical advice
- × No fraud determination
- × No causal medical conclusions
- × No arbitrary Python / SQL / pandas execution

Descriptive healthcare claims analytics ≠ clinical or causal inference

Engineering & Deployment

Same core architecture; different runtime data footprints

Local runtime

Full cleaned Parquet data when available

`data/processed/`

Hugging Face Space

Lightweight public sample data

`data/sample/`

Shared application core

Same planner, registry, analytics engine, LangGraph workflow, response and chart layers.

Clear module boundaries

Data loading • deterministic analytics • bounded planning • approved execution • orchestration • presentation

Validation focus

English/German • single/multi-step • fallback • causal rejection • chart behavior

Portfolio positioning

A controlled applied-DS system demonstrating analytics engineering, orchestration, UX, safety, and deployment.

Python • pandas • Plotly • Gradio • LangGraph • Hugging Face

Medical Insight Explorer Agent — v1.2.0

Controlled multi-step analytics without unrestricted numerical authority

Deterministic

Predefined pandas analytics remain the source of numerical truth.

Controlled

Planner + allowlisted registry constrain what natural language can execute.

Usable

Bilingual personas, guided prompts, transparent workflow status, and live deployment.

[Hugging Face: huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent](https://huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent)

[GitHub: github.com/ArturMelnyk-analyst/Medical-Insight-Explorer-Agent](https://github.com/ArturMelnyk-analyst/Medical-Insight-Explorer-Agent)

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